ARYAN PARAB

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EDUCATION

University of Southern California Los Angeles, CA
M.S. in Computer Science | GPA: 3.85/4.0 January 2025 - December 2026
Coursework: Coursework: Deep Learning and its Applications, Information Retrieval & Web Search Engines, Algorithms, Robotics, Operating Systems, Database Architecture

Thadomal Shahani Engineering College Mumbai, India
Bachelor of Engineering, Computer Engineering | GPA: 9.1/10 August 2019 - May 2023

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C++, C, SQL
AI/ML: PyTorch, TensorFlow, scikit-learn, Transformers, Hugging Face, NLP, Embeddings, Reinforcement Learning, Multimodal AI
Agentic & LLM Systems: Agent Orchestration, Multi-Agent Systems, Tool Calling, RAG, MCP, LangChain, FAISS, Vision LLMs, LLM Evaluation, Prompt Engineering
Backend & Data: FastAPI, Flask, PostgreSQL, MongoDB, SQLite, DuckDB, Pandas, NumPy, REST APIs
Infrastructure: Docker, AWS EC2/S3, Linux, Git, Playwright

EXPERIENCE

Cadence Design Systems San Jose, CA
Generative AI Software Engineering Intern May 2026 – Present
-Architected an autonomous software engineering agent that converts QA specifications into executable test automation through planning, simulation, code generation, execution, debugging, and review.
-Designed a dependency-aware agent planner spanning 1,000+ QA test cases, decomposing requirements into reusable workflows, shared execution dependencies, and coordinated multi-step plans.
-Integrated Playwright simulation with MCP-based Katalon tools, enabling agents to interact with applications, inspect runtime failures, generate targeted fixes, and autonomously retry failed execution steps.
-Built an automated multi-agent Cloudflare security system that ingests daily traffic, filters suspicious requests using WAF and bot scores, clusters activity using JA4 fingerprints and IP signals, and performs agent-driven threat and rule analysis.
-Designed cooperating agents to independently identify threats and diagnose existing Cloudflare rule behavior, with a downstream recommendation agent correlating both analyses to identify coverage gaps and propose new or strengthened security rules.
-Automated the complete security workflow through scheduled daily execution, generating actionable reports containing detected threats, affected traffic, existing rule coverage, rule weaknesses, and recommended remediations for the Cloudflare team.

USC Marshall School of Business Los Angeles, CA
Research Assistant February 2026 - Present
-Built automated LLM evaluation pipelines measuring hallucination, factual consistency, and reasoning quality across models including Claude and Llama using Python and PyTorch.
-Developed benchmarking infrastructure for multi-agent reasoning systems, automating experiment execution, result aggregation, model comparison, and repeated evaluation across configurations.
-Built reusable validation and evaluation workflows to improve experimental reproducibility, identify model failure modes, and systematically compare AI system behavior.

PROJECTS

CNN Article Writer — Preference-Tuned Factual Generation
Llama 3.2, PyTorch, Hugging Face, LoRA, DPO, RAG January 2026
-Fine-tuned Llama 3.2 for factual news generation using a two-stage pipeline with supervised fine-tuning on 13,629 articles followed by Direct Preference Optimization on 500 preference pairs.
-Developed a custom hallucination-evaluation framework and reduced hallucination rates by 86% while preserving structured, professional article generation.
-Applied LoRA with 0.65% trainable parameters and 4-bit quantization, reducing memory requirements by approximately 60% while maintaining sub-2-second inference.

Conversational Database Agent
TypeScript, React, Python, PostgreSQL, MongoDB, Streamlit February 2026-
-Built a multi-agent natural-language database system that converts user requests into validated SQL queries, executes them against DuckDB, and returns explanations and generated visualizations.
-Designed a four-model architecture separating planning, SQL generation, execution, and visualization, allowing specialized models to handle distinct stages of the reasoning pipeline.
-Implemented schema-aware clarification and selective context retrieval to resolve ambiguous requests while reducing unnecessary prompt context.

ACHIEVEMENTS

- Winner - USC Claude AI Hackathon 2025 and USC ACM Hackathon 2026